

Algebra & Equations

ANSWER KEY



Solving linear equations

- 1 Solve $3x - 7 = 11$.
 $3x = 18$, so $x = 6$.
 - 2 Solve $5(x - 2) = 3x + 4$.
 $5x - 10 = 3x + 4 \Rightarrow 2x = 14 \Rightarrow x = 7$.
 - 3 Solve $\frac{3x - 2}{4} = x - 1$.
 $3x - 2 = 4(x - 1) = 4x - 4 \Rightarrow -x = -2 \Rightarrow x = 2$.
 - 4 Solve $2(x + 3) - 5 = 3(x - 1)$.
 $2x + 1 = 3x - 3 \Rightarrow -x = -4 \Rightarrow x = 4$.
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Expanding and simplifying expressions

- 5 Expand and simplify $3(2x - 1) - 2(x - 4)$.
 $6x - 3 - 2x + 8 = 4x + 5$.
 - 6 Expand $(x + 5)(x - 2)$.
 $x^2 - 2x + 5x - 10 = x^2 + 3x - 10$.
 - 7 Expand $(2x - 3)^2$.
 $(2x - 3)^2 = 4x^2 - 12x + 9$.
 - 8 Simplify $\frac{6x^2 - 9x}{3x(2x - 3)}$ (for $x \neq 0$).
 $\frac{6x^2 - 9x}{3x} = 2x - 3$.
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Factoring quadratic expressions

- 9 Factor $x^2 + 7x + 12$.
 $(x + 3)(x + 4)$.
- 10 Factor $x^2 - 2x - 15$.
 $(x - 5)(x + 3)$.
- 11 Factor $2x^2 + 5x - 3$.
 $(2x - 1)(x + 3)$.

- 12 Factor $x^2 - 49$ using the difference of squares.
 $(x - 7)(x + 7)$.
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Solving quadratic equations by factoring

- 13 Solve $x^2 - 5x + 6 = 0$.
 $(x - 2)(x - 3) = 0$, so $x = 2$ or $x = 3$.
- 14 Solve $x^2 + 2x - 8 = 0$.
 $(x + 4)(x - 2) = 0$, so $x = -4$ or $x = 2$.
- 15 Solve $2x^2 - 5x - 3 = 0$.
 $(2x + 1)(x - 3) = 0$, so $x = -\frac{1}{2}$ or $x = 3$.
- 16 Solve $x^2 - 9x + 20 = 0$.
 $(x - 4)(x - 5) = 0$, so $x = 4$ or $x = 5$.
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The quadratic formula and the discriminant

- 17 Use the quadratic formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ to solve $x^2 + 3x + 1 = 0$, leaving the answer in exact form.
 $a = 1, b = 3, c = 1$. $x = \frac{-3 \pm \sqrt{9 - 4}}{2} = \frac{-3 \pm \sqrt{5}}{2}$.
- 18 Find the discriminant of $2x^2 - 4x + 5 = 0$ and state how many real roots it has.
 $\Delta = (-4)^2 - 4(2)(5) = 16 - 40 = -24$. Since $\Delta < 0$, there are no real roots.
- 19 Find the discriminant of $x^2 - 6x + 9 = 0$ and state how many real roots it has.
 $\Delta = (-6)^2 - 4(1)(9) = 36 - 36 = 0$. Since $\Delta = 0$, there is one repeated real root ($x = 3$).
- 20 Use the quadratic formula to solve $3x^2 - 2x - 1 = 0$.
 $\Delta = (-2)^2 - 4(3)(-1) = 4 + 12 = 16$. $x = \frac{2 \pm 4}{6}$, so $x = 1$ or $x = -\frac{1}{3}$.
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Absolute value equations

- 21 Solve $|x - 4| = 9$.
 $x - 4 = 9 \Rightarrow x = 13$, or $x - 4 = -9 \Rightarrow x = -5$.
- 22 Solve $|2x + 3| = 7$.
 $2x + 3 = 7 \Rightarrow x = 2$, or $2x + 3 = -7 \Rightarrow x = -5$.

- 23 Solve $3|x - 1| = 12$.
 $|x - 1| = 4$, so $x - 1 = 4 \Rightarrow x = 5$, or $x - 1 = -4 \Rightarrow x = -3$.
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Linear inequalities

- 24 Solve $2x - 5 < 9$ and graph the solution set on a number line.
 $2x < 14 \Rightarrow x < 7$.
- 25 Solve $-3x + 4 \geq -8$.
 $-3x \geq -12$. Dividing by -3 reverses the inequality: $x \leq 4$.
- 26 Solve the compound inequality $5 < 2x + 1 \leq 11$.
Subtract 1: $4 < 2x \leq 10$. Divide by 2: $2 < x \leq 5$.
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Systems of linear equations

- 27 Solve the system $x + y = 10$, $x - y = 2$.
Adding the equations: $2x = 12 \Rightarrow x = 6$. Then $y = 10 - 6 = 4$.
- 28 Solve the system $3x + 2y = 16$, $x - y = 2$.
From the second equation, $x = 2 + y$. Substituting: $3(2 + y) + 2y = 16 \Rightarrow 6 + 5y = 16 \Rightarrow y = 2$. Then $x = 4$.
- 29 This question has two parts. The sum of two numbers is 15 and their difference is 3.
- a Write a system of two equations for the numbers x and y . $x + y = 15$ and $x - y = 3$.
- b Solve the system to find both numbers. Adding the equations: $2x = 18 \Rightarrow x = 9$. Then $y = 15 - 9 = 6$.
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Rational equations

- 30 Solve $\frac{x}{x-2} = 3$.
 $x = 3(x-2) = 3x - 6 \Rightarrow -2x = -6 \Rightarrow x = 3$.
- 31 Solve $\frac{2}{x+1} = \frac{1}{3}$.
Cross-multiplying: $6 = x + 1 \Rightarrow x = 5$.
- 32 Solve $\frac{x+1}{x-1} = 2$.
 $x + 1 = 2(x - 1) = 2x - 2 \Rightarrow -x = -3 \Rightarrow x = 3$.
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Solving for a variable in a formula

- 33 The area of a triangle is $A = \frac{1}{2}bh$. Solve for h .
$$h = \frac{2A}{b}.$$

- 34 The circumference of a circle is $C = 2\pi r$. Solve for r .
$$r = \frac{C}{2\pi}.$$

- 35 The formula $v = v_0 + at$ gives velocity under constant acceleration. Solve for t .
$$t = \frac{v - v_0}{a}.$$
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Linear-quadratic systems

- 36 Solve the system $y = x^2 - 1$, $y = 2x + 2$.
Setting equal: $x^2 - 1 = 2x + 2 \Rightarrow x^2 - 2x - 3 = 0 \Rightarrow (x - 3)(x + 1) = 0$. So $x = 3$, $y = 8$ or $x = -1$, $y = 0$.
- 37 Solve the system $y = x^2$, $y = 3x - 2$.
Setting equal: $x^2 = 3x - 2 \Rightarrow x^2 - 3x + 2 = 0 \Rightarrow (x - 1)(x - 2) = 0$. So $x = 1$, $y = 1$ or $x = 2$, $y = 4$.
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Word problems: setting up and solving equations

- 38 This question has two parts. A rectangular garden has a length that is 4 meters more than its width w . Its area is 96 square meters.
- a Write an equation relating w to the area. $w(w + 4) = 96$, i.e. $w^2 + 4w - 96 = 0$.
 - b Solve for w , and state the garden's length. Factoring: $(w - 8)(w + 12) = 0$. Since width is positive, $w = 8$ meters, so the length is $w + 4 = 12$ meters.
- 39 This question has two parts. A mobile-data plan charges a fixed monthly fee f plus a rate r per gigabyte used. One month, 5 GB of data cost a total of 65 dollars; another month, 9 GB cost a total of 93 dollars.
- a Write a system of two equations for f and r . $f + 5r = 65$ and $f + 9r = 93$.
 - b Solve the system to find the fixed fee and the rate per gigabyte. Subtracting the first equation from the second: $4r = 28 \Rightarrow r = 7$. Then $f = 65 - 5(7) = 30$. Fixed fee: 30 dollars; rate: 7 dollars per GB.
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MCQ — Solving linear equations

40 Solve $4x - 9 = 15$.

a. $x = 6$ ✓

b. $x = -6$

c. $x = 24$

d. $x = 1.5$

$$4x = 24 \Rightarrow x = 6.$$

41 Solve $6(x - 3) = 2x + 10$.

a. $x = -7$

b. $x = 4$

c. $x = 7$ ✓

d. $x = 1$

$$6x - 18 = 2x + 10 \Rightarrow 4x = 28 \Rightarrow x = 7.$$

42 Solve $\frac{2x - 1}{3} = x - 2$.

a. $x = 5$ ✓

b. $x = 1$

c. $x = 7$

d. $x = -5$

$$2x - 1 = 3x - 6 \Rightarrow x = 5.$$

43 Solve $3(x + 4) - 7 = 2(x - 1)$.

a. $x = -7$ ✓

b. $x = -3$

c. $x = 3$

d. $x = 7$

$$3x + 5 = 2x - 2 \Rightarrow x = -7.$$

44 Solve $5x + 3 = 2x - 9$.

a. $x = -2$

b. $x = 2$

c. $x = 4$

d. $x = -4$ ✓

$$3x = -12 \Rightarrow x = -4.$$

MCQ — Expanding and simplifying expressions

45 Expand and simplify $4(3x - 2) - 3(x - 5)$.

a. $15x - 23$

b. $9x + 23$

c. $9x + 7$ ✓

d. $9x - 23$

$$12x - 8 - 3x + 15 = 9x + 7.$$

46 Expand $(x + 7)(x - 3)$.

a. $x^2 + 10x - 21$

b. $x^2 - 4x - 21$

c. $x^2 + 4x - 21$ ✓

d. $x^2 + 4x + 21$

$$(x + 7)(x - 3) = x^2 + 4x - 21.$$

47 Expand $(3x - 1)^2$.

a. $3x^2 - 6x + 1$

b. $9x^2 - 1$

c. $9x^2 - 6x + 1$ ✓

d. $9x^2 - 6x - 1$

$$(3x - 1)^2 = 9x^2 - 6x + 1.$$

48 Simplify $\frac{8x^2 - 12x}{4x}$ (for $x \neq 0$).

a. $2x - 12$

b. $8x - 3$

c. $2x - 3$ ✓

d. $2x^2 - 3$

$$\frac{4x(2x - 3)}{4x} = 2x - 3.$$

MCQ — Factoring quadratic expressions

49 Factor $x^2 + 9x + 14$.

a. $(x + 2)(x + 7)$ ✓

b. $(x - 2)(x - 7)$

c. $(x + 1)(x + 14)$

d. $(x + 14)(x - 1)$

Need two numbers multiplying to 14, adding to 9: 2, 7.

50 Factor $x^2 - 3x - 18$.

- a. $(x - 9)(x + 2)$
- b. $(x - 2)(x + 9)$
- c. $(x - 6)(x + 3)$ ✓
- d. $(x + 6)(x - 3)$

Need two numbers multiplying to -18 , adding to -3 : $-6, 3$.

51 Factor $3x^2 + 7x + 2$.

- a. $(3x + 1)(x + 2)$ ✓
- b. $(3x - 1)(x - 2)$
- c. $(3x + 2)(x + 1)$
- d. $(x + 1)(3x + 2)$

$$(3x + 1)(x + 2) = 3x^2 + 6x + x + 2 = 3x^2 + 7x + 2.$$

52 Factor $x^2 - 81$ using the difference of squares.

- a. $(x - 9)^2$
- b. $(x - 9)(x + 9)$ ✓
- c. $(x + 9)^2$
- d. $(x - 81)(x + 1)$

$$x^2 - 9^2 = (x - 9)(x + 9).$$

53 Factor $2x^2 - x - 6$.

- a. $(2x + 3)(x - 2)$ ✓
- b. $(2x - 3)(x + 2)$
- c. $(x + 3)(2x - 2)$
- d. $(2x + 1)(x - 6)$

$$(2x + 3)(x - 2) = 2x^2 - 4x + 3x - 6 = 2x^2 - x - 6.$$

MCQ — Solving quadratic equations by factoring

54 Solve $x^2 - 7x + 10 = 0$.

a. $x = 1$ or $x = 10$

b. $x = 7$ or $x = 10$

c. $x = -2$ or $x = -5$

d. $x = 2$ or $x = 5$ ✓

$$(x - 2)(x - 5) = 0 \Rightarrow x = 2 \text{ or } x = 5.$$

55 Solve $x^2 + 3x - 10 = 0$.

a. $x = -3$ or $x = 10$

b. $x = -10$ or $x = 1$

c. $x = 5$ or $x = -2$

d. $x = -5$ or $x = 2$ ✓

$$(x + 5)(x - 2) = 0 \Rightarrow x = -5 \text{ or } x = 2.$$

56 Solve $3x^2 - 11x - 4 = 0$.

a. $x = -\frac{1}{3}$ or $x = 4$ ✓

b. $x = \frac{1}{3}$ or $x = -4$

c. $x = 1$ or $x = 4$

d. $x = -\frac{1}{3}$ or $x = -4$

$$(3x + 1)(x - 4) = 0 \Rightarrow x = -\frac{1}{3} \text{ or } x = 4.$$

57 Solve $x^2 - 11x + 24 = 0$.

a. $x = 1$ or $x = 24$

b. $x = 4$ or $x = 6$

c. $x = -3$ or $x = -8$

d. $x = 3$ or $x = 8$ ✓

$(x - 3)(x - 8) = 0 \Rightarrow x = 3$ or $x = 8$.

MCQ — The quadratic formula and the discriminant

58 Use the quadratic formula to solve $x^2 + 5x + 2 = 0$, leaving the answer in surd form.

a. $x = \frac{5 \pm \sqrt{17}}{2}$

b. $x = \frac{-5 \pm \sqrt{17}}{4}$

c. $x = \frac{-5 \pm \sqrt{33}}{2}$

d. $x = \frac{-5 \pm \sqrt{17}}{2}$ ✓

Discriminant $= 25 - 8 = 17$; $x = \frac{-5 \pm \sqrt{17}}{2}$.

59 Find the discriminant of $3x^2 - 6x + 5 = 0$ and state how many real roots it has.

a. 96; two real roots

b. 24; two real roots

c. -24; no real roots ✓

d. -24; one real root

$(-6)^2 - 4(3)(5) = 36 - 60 = -24 < 0$: no real roots.

60 Find the discriminant of $x^2 - 8x + 16 = 0$ and state how many real roots it has.

a. 0; one real root ✓

b. 64; two real roots

c. 0; two real roots

d. -64; no real roots

$(-8)^2 - 4(1)(16) = 64 - 64 = 0$: one repeated real root.

61 Use the quadratic formula to solve $2x^2 + 3x - 2 = 0$.

a. $x = 1$ or $x = -2$

b. $x = -0.5$ or $x = 2$

c. $x = 0.5$ or $x = -2$ ✓

d. $x = 0.5$ or $x = 2$

Discriminant $= 9 + 16 = 25$; $x = \frac{-3 \pm 5}{4}$: $x = 0.5$ or $x = -2$.

MCQ — Absolute value equations

62 Solve $|x - 6| = 11$.

a. $x = -17$ or $x = 5$

b. $x = 5$ or $x = -17$

c. $x = 17$ or $x = -5$ ✓

d. $x = 17$ or $x = 5$

$x - 6 = 11$ or $x - 6 = -11$: $x = 17$ or $x = -5$.

63 Solve $|3x + 2| = 13$.

a. $x = \frac{11}{3}$ or $x = 5$

b. $x = 5$ or $x = -5$

c. $x = -\frac{11}{3}$ or $x = 5$

d. $x = \frac{11}{3}$ or $x = -5$ ✓

$3x + 2 = 13 \Rightarrow x = \frac{11}{3}; 3x + 2 = -13 \Rightarrow x = -5.$

64 Solve $4|x - 2| = 20$.

a. $x = 7$ or $x = -3$ ✓

b. $x = -7$ or $x = 3$

c. $x = 5$ or $x = -5$

d. $x = 7$ or $x = 3$

$|x - 2| = 5 \Rightarrow x - 2 = 5$ or $x - 2 = -5; x = 7$ or $x = -3.$

65 Solve $|2x - 5| = 9$.

a. $x = 7$ or $x = 2$

b. $x = -7$ or $x = 2$

c. $x = 2$ or $x = -7$

d. $x = 7$ or $x = -2$ ✓

$2x - 5 = 9 \Rightarrow x = 7; 2x - 5 = -9 \Rightarrow x = -2.$

66 The graph shows $y = |x - 3|$. At what x -value does the graph reach its minimum?

a. $x = 1$

b. $x = 3$ ✓

c. $x = -3$

d. $x = 0$

The vertex of the V-shape (the minimum) is at $x = 3$.

MCQ — Linear inequalities

67 Solve $3x - 7 < 14$, and identify the graph of its solution set.

a. $x < 7$ ✓

b. $x \geq 7$

c. $x \leq 7$

d. $x > 7$

$$3x < 21 \Rightarrow x < 7.$$

68 Solve $-4x + 6 \geq -10$, and identify the graph of its solution set.

a. $x > 4$

b. $x < 4$

c. $x \leq 4$ ✓

d. $x \geq 4$

$$-4x \geq -16; \text{ dividing by } -4 \text{ flips the sign: } x \leq 4.$$

69 Solve the compound inequality $7 < 3x + 1 \leq 19$.

a. $2 < x \leq 6$ ✓

b. $2 < x < 6$

c. $2 \leq x < 6$

d. $6 < x \leq 2$

$$6 < 3x \leq 18 \Rightarrow 2 < x \leq 6.$$

70 Solve $2x + 5 > -3$ and $x - 1 < 6$ together (both must hold). Find the combined solution set.

a. $x > -4$

b. $-4 < x \leq 7$

c. $x < 7$

d. $-4 < x < 7$ ✓

$2x + 5 > -3 \Rightarrow x > -4$; $x - 1 < 6 \Rightarrow x < 7$. Together: $-4 < x < 7$.

MCQ — Systems of linear equations

71 Solve the system $x + y = 14$, $x - y = 4$.

a. $x = 5, y = 9$

b. $x = 10, y = 4$

c. $x = 9, y = 5$ ✓

d. $x = 9, y = 4$

Adding: $2x = 18 \Rightarrow x = 9$; then $y = 5$.

72 Solve the system $4x + 3y = 26$, $x - y = 3$.

a. $x = 5, y = 2$ ✓

b. $x = 2, y = 5$

c. $x = 8, y = 5$

d. $x = 5, y = 3$

$x = y + 3 \Rightarrow 4(y + 3) + 3y = 26 \Rightarrow 7y = 14 \Rightarrow y = 2$; $x = 5$.

73 The sum of two numbers is 22 and their difference is 6. Find the two numbers.

- a. 11 and 11
- b. 14 and 6
- c. 16 and 6
- d. 14 and 8 ✓

$$x + y = 22, x - y = 6 \Rightarrow x = 14, y = 8.$$

74 Solve the system $2x - y = 7$, $3x + 2y = 21$.

- a. $x = 5, y = -3$
- b. $x = 5, y = 3$ ✓
- c. $x = 3, y = 5$
- d. $x = 7, y = 7$

$$y = 2x - 7 \Rightarrow 3x + 2(2x - 7) = 21 \Rightarrow 7x = 35 \Rightarrow x = 5; y = 3.$$

MCQ — Rational equations

75 Solve $\frac{x}{x-3} = 4$.

- a. $x = 12$
- b. $x = -4$
- c. $x = 4$ ✓
- d. $x = 3$

$$x = 4(x - 3) = 4x - 12 \Rightarrow -3x = -12 \Rightarrow x = 4.$$

76 Solve $\frac{3}{x+2} = \frac{1}{4}$.

a. $x = 6$

b. $x = 14$

c. $x = 10$ ✓

d. $x = -10$

Cross-multiply: $12 = x + 2 \Rightarrow x = 10$.

77 Solve $\frac{x+2}{x-2} = 3$.

a. $x = 2$

b. $x = 8$

c. $x = 4$ ✓

d. $x = -4$

$x + 2 = 3(x - 2) = 3x - 6 \Rightarrow 8 = 2x \Rightarrow x = 4$.

78 Solve $\frac{4}{x-1} = \frac{2}{x+2}$.

a. $x = -5$ ✓

b. $x = 5$

c. $x = -1$

d. $x = 2$

Cross-multiply: $4(x + 2) = 2(x - 1) \Rightarrow 4x + 8 = 2x - 2 \Rightarrow x = -5$.

79 Solve $\frac{6}{x} = \frac{2}{x-4}$.

a. $x = 12$

b. $x = 4$

c. $x = -6$

d. $x = 6$ ✓

Cross-multiply: $6(x - 4) = 2x \Rightarrow 6x - 24 = 2x \Rightarrow x = 6$.

MCQ — Solving for a variable in a formula

80 The perimeter of a rectangle is $P = 2l + 2w$. Solve for l .

a. $l = \frac{P}{2} - w$

b. $l = \frac{P - 2w}{2}$ ✓

c. $l = \frac{P + 2w}{2}$

d. $l = P - 2w$

$$P - 2w = 2l \Rightarrow l = \frac{P - 2w}{2} \text{ (equivalently } \frac{P}{2} - w \text{)}.$$

81 The formula for simple interest is $I = Prt$. Solve for r .

a. $r = \frac{I}{P}$

b. $r = \frac{I}{Pt}$ ✓

c. $r = \frac{Pt}{I}$

d. $r = I - Pt$

$$\text{Dividing both sides by } Pt: r = \frac{I}{Pt}.$$

82 The formula $F = \frac{9}{5}C + 32$ converts Celsius to Fahrenheit. Solve for C .

a. $C = \frac{5(F - 32)}{9}$ ✓

b. $C = \frac{5F - 32}{9}$

c. $C = \frac{5}{9}F + 32$

d. $C = \frac{9(F - 32)}{5}$

$$F - 32 = \frac{9}{5}C \Rightarrow C = \frac{5(F - 32)}{9}.$$

83 The volume of a cylinder is $V = \pi r^2 h$. Solve for h .

a. $h = \frac{V}{\pi r^2}$ ✓

b. $h = V - \pi r^2$

c. $h = \frac{V}{\pi r}$

d. $h = \frac{\pi r^2}{V}$

Dividing both sides by πr^2 : $h = \frac{V}{\pi r^2}$.

MCQ — Linear-quadratic systems

84 Solve the system $y = x^2 - 4$, $y = 3x$.

a. $(-4, -12)$ and $(1, 3)$

b. $(-1, -3)$ and $(1, 3)$

c. $(4, 12)$ and $(1, 3)$

d. $(4, 12)$ and $(-1, -3)$ ✓

$$x^2 - 4 = 3x \Rightarrow x^2 - 3x - 4 = 0 \Rightarrow (x - 4)(x + 1) = 0: x = 4, y = 12 \text{ or } x = -1, y = -3.$$

85 Solve the system $y = x^2$, $y = 2x + 3$.

a. $(-3, 9)$ and $(1, 1)$

b. $(3, 9)$ and $(1, 1)$

c. $(-1, 1)$ and $(1, 1)$

d. $(3, 9)$ and $(-1, 1)$ ✓

$$x^2 - 2x - 3 = 0 \Rightarrow (x - 3)(x + 1) = 0: x = 3, y = 9 \text{ or } x = -1, y = 1.$$

86 The graph shows the system $y = x^2 - 4$, $y = 3x$. What are the points of intersection?

- a. $(4, 12)$ and $(1, 3)$
- b. $(-4, -12)$ and $(1, 3)$
- c. $(-1, -3)$ and $(1, 3)$
- d. $(4, 12)$ and $(-1, -3)$ ✓

The curve and line cross at $(-1, -3)$ and $(4, 12)$.

87 The graph shows the system $y = x^2$, $y = 2x + 3$. What are the points of intersection?

- a. $(3, 9)$ and $(1, 1)$
- b. $(-1, 1)$ and $(1, 1)$
- c. $(-3, 9)$ and $(1, 1)$
- d. $(3, 9)$ and $(-1, 1)$ ✓

The curve and line cross at $(-1, 1)$ and $(3, 9)$.

MCQ — Word problems: setting up and solving equations

88 A rectangular garden has a length that is 5 meters more than its width, and its area is 84 square meters. Find the width.

- a. 12
- b. 7 ✓
- c. 9
- d. 14

$w(w + 5) = 84 \Rightarrow (w + 12)(w - 7) = 0$; the positive root is $w = 7$.

89 A mobile-data plan charges a fixed monthly fee of 40 riyals plus 2 riyals per GB used. If a user's bill was 68 riyals, how many GB did they use?

a. 34

b. 14 ✓

c. 54

d. 28

$$40 + 2g = 68 \Rightarrow 2g = 28 \Rightarrow g = 14.$$